Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech (EE) Theory Examination

May 2018

EE308.01/EE308 High Voltage Engineering

Date: 05.05.2018, Saturday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1.a Explain the mechanism for the occurrence of back-flashover. **[04]**

b. Write the functions of followings: **[02]**
(1) Corona Ring & (2) Arcing Horns

Q – 2.a Discuss the influence of space charge over the breakdown voltage. **[03]**

Q – 2.b **Attempt any two questions.** **[12]**

(i) Attempt the followings:
1. Differentiate positive and negative corona discharges.
2. Why the dielectric strength of electronegative gases is higher than electropositive gas?
3. Define the terms: Uniform and Non-uniform field.

(ii) Discuss the various classifications of liquid dielectrics.

(iii) Explain the mechanism of breakdown due to treeing and tracking phenomena in solid dielectrics.

Q - 3 **Attempt any two questions.** **[14]**

a. Describe the current growth phenomenon in a gas subjected to uniform electric fields and derive the condition for breakdown obtained in a gas.

b. Explain the mechanism of breakdown due to internal discharges in solid dielectrics.

c. Draw the layout of oil purification system.

SECTION – II

Q - 4 List out the various methods used for the measurement of high voltages and currents. **[04]**

Q – 5.a A 100 kVA, 400 V/250 kV testing transformer has 8% leakage reactance and 2% resistance on 100 kVA base. A cable has to be tested at 500 kV using the above transformer as a resonant transformer at 50 Hz. If the charging current of the cable at 500 kV is 0.4 A, find the series inductance required. Assume 2% resistance for the inductor and 2% resistance for the connecting leads. Neglect the dielectric losses of the cable. Also determine the input voltage and input power required to the transformer. **[08]**

OR

Q – 5.a Explain voltage doubler circuit with output waveforms. Also derive the equation for ripple voltage magnitude. **[08]**

Q – 5.b Draw the impulse voltage wave for lightning overvoltage and switching overvoltage. Also mention the standard values with tolerances for lightning and switching impulses. **[04]**

Q – 5.c Explain electrostatic voltmeter in detail. **[08]**

OR

Q – 5.c A Cockcroft – Walton type voltage multiplier has eight stages with capacitances all equal to 0.05 µF. The secondary voltage of transformer is 125 kV. If the load current to be supplied is 5 mA, find (a) % Ripple, (b) % Regulation, (c) Optimum number of stages & (d) Maximum output voltage at when frequency of supply is 150 Hz. **[08]**

Q – 6.a An Impulse generator has 12 capacitors of 0.12µF and 200 kV rating. The wave front and wave tail resistances are 1.25 kΩ and 4 kΩ respectively. If the load capacitance including the test object is 1000 PF, find the wave front time and wave tail time. **[04]**

Q – 6.b Draw the layout of short circuit plant used for testing of circuit breakers. **[07]**

OR

Q – 6.b Derive the equation for energy associated with single discharge. **[07]**

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